

1. Final Structure

```
amaanitvam-platform/  
|  
|-- apps/  
| |-- website/  
| | |-- public/  
| | | |-- images/  
| | | |-- documents/  
| | | |-- icons/  
| | | `-- favicon.ico  
| | |  
| | |-- src/  
| | | |-- components/  
| | | |-- pages/  
| | | |-- styles/  
| | | |-- scripts/  
| | | `-- config/  
| | |  
| | |-- package.json  
| | |-- vite.config.js  
| | `-- .env.example  
| |  
|-- admin-portal/  
| |-- public/  
| |-- src/  
| | |-- assets/  
| | |-- components/  
| | |-- features/  
| | |-- layouts/  
| | |-- routes/  
| | |-- services/  
| | |-- contexts/  
| | |-- hooks/  
| | |-- utils/  
| | |-- styles/  
| | |-- App.jsx  
| | `-- main.jsx
```

```
| | |-- package.json
| | |-- vite.config.js
| | `-- .env.example
| |
| |-- dashboard/
| | |-- public/
| | |-- src/
| | | |-- assets/
| | | |-- components/
| | | |-- features/
| | | |-- layouts/
| | | |-- routes/
| | | |-- services/
| | | |-- contexts/
| | | |-- hooks/
| | | |-- utils/
| | | |-- styles/
| | | |-- App.jsx
| | | `-- main.jsx
| |-- package.json
| |-- vite.config.js
| `-- .env.example
|
|-- server/
| |-- src/
| | |-- config/
| | |-- modules/
| | |-- middleware/
| | |-- shared/
| | |-- services/
| | |-- jobs/
| | |-- app.js
| | `-- server.js
| |-- scripts/
| |-- tests/
| |-- package.json
| `-- .env.example
|
|-- packages/
| |-- ui/
```

```
| |-- shared-utils/  
| `-- shared-config/  
|  
|-- docs/  
|-- .github/  
|-- .editorconfig  
|-- .gitignore  
|-- package.json  
|-- README.md  
`-- LICENSE
```

2. Public Website Structure

```
apps/website/  
|-- public/  
| |-- images/  
| | |-- common/  
| | |-- programs/  
| | |-- gallery/  
| | |-- webinars/  
| | `-- team/  
|-- documents/  
| |-- annual-reports/  
| |-- policies/  
| | `-- certificates/  
| `-- icons/  
|  
`-- src/  
    |-- components/  
    | |-- navbar.html  
    | |-- footer.html  
    | |-- social-bar.html  
    | `-- donation-cta.html  
    |-- pages/  
    | |-- index.html  
    | |-- about.html  
    | |-- impact.html  
    | |-- programs.html  
    | |-- gallery.html  
    | |-- volunteer.html  
    | |-- internship.html  
    | |-- contact.html  
    | |-- collaborations.html  
    | |-- faq.html  
    | |-- certificate-verification.html  
    | |-- digital-library.html  
    | |-- courses.html  
    | `-- webinars-competitions.html  
    |-- styles/  
    | |-- variables.css  
    | |-- reset.css
```

```
| |-- base.css
| |-- layout.css
| |-- components.css
| |-- responsive.css
| `-- pages/
|-- scripts/
|   |-- app.js
|   |-- api-client.js
|   |-- component-loader.js
|   |-- navbar.js
|   |-- gallery.js
|   |-- forms.js
|   |-- library.js
|   `-- courses.js
```

3. Admin Portal Structure

```
apps/admin-portal/src/  
|-- assets/  
|-- components/  
| |-- common/  
| |-- navigation/  
| |-- forms/  
| |-- tables/  
| |-- modals/  
| |-- feedback/  
| `-- guards/  
|-- features/  
| |-- auth/  
| |-- dashboard/  
| |-- candidates/  
| |-- members/  
| |-- departments/  
| |-- tasks/  
| |-- meetings/  
| |-- announcements/  
| |-- donations/  
| |-- certificates/  
| |-- gallery/  
| |-- cms/  
| |-- digital-library/  
| |-- courses/  
| |-- reports/  
| `-- settings/  
|-- layouts/  
|-- routes/  
|-- services/  
|-- contexts/  
|-- hooks/  
|-- utils/  
|-- styles/  
|-- App.jsx  
`-- main.jsx
```

4. Team Dashboard Structure

```
apps/dashboard/src/  
|-- assets/  
|-- components/  
| |-- common/  
| |-- navigation/  
| |-- tables/  
| |-- forms/  
| `-- guards/  
|-- features/  
| |-- auth/  
| |-- home/  
| |-- profile/  
| |-- tasks/  
| |-- projects/  
| |-- meetings/  
| |-- announcements/  
| |-- departments/  
| |-- reports/  
| `-- attendance/  
|-- layouts/  
|-- routes/  
|-- services/  
|-- contexts/  
|-- hooks/  
|-- utils/  
|-- styles/  
|-- App.jsx  
`-- main.jsx
```

5. Backend Architecture

```
server/src/  
|-- config/  
| |-- database.js  
| |-- firebase.js  
| |-- clouinary.js  
| |-- razorpay.js  
| |-- mailer.js  
| `-- env.js  
|  
|-- modules/  
| |-- auth/  
| |-- users/  
| |-- candidates/  
| |-- members/  
| |-- departments/  
| |-- tasks/  
| |-- meetings/  
| |-- announcements/  
| |-- projects/  
| |-- donations/  
| |-- certificates/  
| |-- gallery/  
| |-- cms/  
| |-- digital-library/  
| |-- courses/  
| |-- volunteers/  
| |-- internships/  
| |-- reports/  
| `-- notifications/  
|  
|-- middleware/  
| |-- authenticate.js  
| |-- authorize.js  
| |-- validate.js  
| |-- upload.js  
| |-- rateLimiter.js  
| |-- errorHandler.js  
| `-- notFound.js
```



```
|  
|-- shared/  
| |-- constants/  
| |-- errors/  
| |-- logger/  
| |-- response/  
| `-- utils/  
|  
|-- services/  
| |-- email.service.js  
| |-- sms.service.js  
| |-- storage.service.js  
| `-- activity.service.js  
|  
|-- jobs/  
|-- app.js  
`-- server.js
```

6. Standard Backend Module Pattern

```
modules/certificates/  
|-- certificate.model.js  
|-- certificate.controller.js  
|-- certificate.service.js  
|-- certificate.routes.js  
|-- certificate.validation.js  
|-- certificate.repository.js  
`-- certificate.utils.js
```

File	Responsibility
Model	Database schema and indexes
Controller	HTTP request and response handling
Service	Business logic
Routes	Endpoint definitions and middleware chain
validation	Request payload validation
repository	Database queries
utils	Feature-specific helper functions

7. Standard React Feature Pattern

```
features/certificates/  
|-- pages/  
| `-- CertificatesPage.jsx  
|-- components/  
| |-- CertificateTable.jsx  
| |-- CertificateForm.jsx  
| `-- CertificateDetailsModal.jsx  
|-- services/  
| `-- certificateService.js  
|-- hooks/  
| `-- useCertificates.js  
|-- utils/  
| `-- certificateHelpers.js  
`-- index.js
```

8. Digital Library Module

server/src/modules/digital-library/

|-- libraryItem.model.js

|-- libraryCategory.model.js

|-- library.controller.js

|-- library.service.js

|-- library.repository.js

|-- library.routes.js

|-- library.validation.js

`-- library.constants.js

apps/admin-portal/src/features/digital-library/

|-- pages/DigitalLibraryPage.jsx

|-- components/LibraryTable.jsx

|-- components/LibraryItemForm.jsx

|-- components/CategoryManager.jsx

|-- components/PublishToggle.jsx

|-- services/libraryService.js

`-- hooks/useLibraryItems.js

apps/website/src/pages/digital-library.html

apps/website/src/scripts/library.js

apps/website/src/styles/pages/library.css

9. Courses Module

server/src/modules/courses/

|-- course.model.js

|-- lesson.model.js

|-- enrollment.model.js

|-- course.controller.js

|-- course.service.js

|-- course.repository.js

|-- course.routes.js

`-- course.validation.js

apps/admin-portal/src/features/courses/

|-- pages/CoursesPage.jsx

|-- components/CourseForm.jsx

|-- components/CourseTable.jsx

|-- components/LessonBuilder.jsx

|-- components/PublishSettings.jsx

`-- services/courseService.js

10. Root Workspace Configuration

```
{
  "name": "amaanitvam-
platform", "private": true,
  "workspac
es": [
    "apps/*",
    "server",
    "packages
/*"
  ],
  "scripts": {
    "dev:website": "npm run dev --workspace
apps/website", "dev:admin": "npm run dev --
workspace apps/admin-portal",
    "dev:dashboard": "npm run dev --workspace
apps/dashboard", "dev:server": "npm run dev --
workspace server",
    "build": "npm run build --workspaces
--if-present", "test": "npm run test --
workspaces --if-present"
  }
}
```

11. Final .gitignore

```
#
Dependenc
ies
node_modu
les/
**/node_modules/

#
Builds
dist/
build/
**/dist/
**/build/

# Environment files
.env
.env.*
**/.env
**/.env.*
!.env.example
!**/.env.example

# Runtime
uploads
uploads/
server/uploads/

# Logs
*.l
og
log
s/

# Test
coverage
coverage/
**/coverage/
```

```
# OS and editors
.DS_Store
Thumbs.db
.vscode/*
!.vscode/extensions.json
.idea/

# Temporary
files tmp/
temp/
*.tmp
```

12. Naming Conventions

Type	Convention
Folders	kebab-case
React components	PascalCase.jsx
Hooks	useSomething.js

Backend files	feature.type.js
Variables/functions	camelCase
Constants	UPPER_SNAKE_CASE

13. Roles and Access Control

- super_admin
- admin
- coordinator
- faculty
- team_member
- intern
- content_editor
- viewer

Frontend visibility is not security. Every protected backend endpoint must authenticate the user and verify the required role or permission.

```
router.post(
  '/',
  authenticate,
  authorize(["super_admin", "admin", "content_editor"]),
  validate(createCourseSchema),
  createCourse
);
```

14. Safe Migration Plan

Phase 1 - Cleanup

- Create a stable Git tag before restructuring.
- Fix the root .gitignore.
- Add .env.example files.
- Remove node_modules and dist from Git tracking.
- Standardise filenames and remove duplicate assets.

Phase 2 - Create New Top-Level Folders

- Create apps, server, packages, and docs.
- Copy code first; do not immediately delete working files.

Phase 3 - Move Frontends

- Move Website to apps/website.
- Move admin portal to apps/admin-portal.
- Move dashboard to apps/dashboard.
- Run and build each application separately.

Phase 4 - Move Backend

- Move backend to server.
- First preserve the existing structure inside server/src.
- Migrate modules gradually.
- Start with gallery, certificates, digital library, and courses.
- Move authentication last because both portals depend on it.

Phase 5 - Testing

- Website loading and navigation
- Admin and dashboard login
- Firebase token verification
- API calls and CORS
- Certificate verification and download
- Donation flow
- Gallery upload
- Forms and mobile navigation
- Production builds

Phase 6 - Deployment Roots

- Website: apps/website
- Admin: apps/admin-portal
- Dashboard: apps/dashboard
- Backend: server